Strings in C

Strings

Definition: -

- A String in C programming is a sequence of characters terminated with a null character ‘\0’.
- The C String is stored as an array of characters.
- The difference between a character array and a C string is that the string in C is terminated with a unique character ‘\0’

Declaration Syntax

char *string_name*[*size*];

Where,

string_name - any name given to the string variable

size - length of the string, i.e the number of characters strings will store.

Strings

string by char array

```
char ch[9]={‘C', ‘o', ‘m', ‘p', ‘u', ‘t', ‘e', ‘r','\0'};
```

string by the string literal

```
char ch[]=“computer";
```

Difference between char array and string literal

- There are two main differences between char array and literal.
- We need to add the null character '\0' at the end of the array by ourself whereas, it is appended internally by the compiler in the case of the character array.
- The string literal cannot be reassigned to another set of characters whereas, we can reassign the characters of the array.

Initializing String

An extra terminating character - ***Null character ('\0') used to indicate the termination of a string that differs strings from normal character arrays***.

Initializing a String

- **Assigning a String Literal without Size**

  char str[] = "Computer";

- **Assigning a String Literal with a Predefined Size**

  char str[50] = "Computer";

Note: *When a Sequence of characters enclosed in the double quotation marks is encountered by the compiler, a null character '\0' is appended at the end of the string by default.*

Important : Always account for one extra space which will be assigned to the null character. If we want to store a string of size n then we should always declare a string with a size equal to or greater than n+1.

Initializing String

- **Assigning Character by Character with Size**

  char str[14] = { ‘C',‘o',‘m',‘p',‘u',‘t',‘e','r','**0**'};

- **Assigning Character by Character without Size**

  char str[ ] = { ‘C',‘o',‘m',‘p',‘u',‘t',‘e','r','**0**'};

Printing String

```
#include <stdio.h>
#include <string.h>
int main()
{
  // declare and initialize string
  char str[] = “Computer";
  // print string
  printf("%s\n", str);
  int length = 0;
  length = strlen(str);
  // displaying the length of string
  printf("Length of string str is %d", length);
  return 0;
}
```

Note: *The C language provides an access specifier “**%s**” which can be used to print and read strings directly.*

Read a String Input From the User

```
#include<stdio.h>
int main()
{
   // declaring string
   char str[50];
   scanf("%s",str); // reading string
   printf("%s",str); // print string
   return 0;
}
```

The ‘&’ sign is used to provide the address of the variable to the scanf() function to store the value read in memory.

As str[] is a character array so using str without braces ‘[‘ and ‘]’ will give the base address of this string.

That’s why we have not used ‘&’ in this case as we are already providing the base address of the string to scanf.

Note: The string is read only till the whitespace is encountered.

Read a String Input with whitespaces from the User

```
// C Program to take string separated by whitespace using scanset characters
#include <stdio.h>
int main()
{
  char str[20];
  scanf("%[^\n]s", str); // using scanset in scanf

  printf("%s", str); // printing read string
  return 0;
}
```